

Python Programming Test Paper

Section A

Q1. Login + Odd Digit Counter

Write a program that allows only 3 login attempts.
Correct credentials: username = 'admin', password = '1234'.
If login successful, ask the user to enter a number and display the count of odd digits.
If all attempts fail, print 'Account Locked'.

Q2. Student Pass Checker

Take marks of 5 subjects and validate marks (0–100). Passing marks per subject = 40.
If any mark is invalid, print 'Invalid Marks'.
If student passes all subjects, calculate percentage and print:
Excellent (≥ 80), Good (≥ 60), Average (≥ 40).
Else print 'Backlog in subjects'.

Q3. Smart Calculator

Create a menu-driven calculator using while loop.
Menu: 1 → Addition, 2 → Multiplication, 3 → Division, 4 → Exit.
Use functions and handle division by zero.

Section B

Q4. Number Property Checker

Take a number and display total digits, sum of digits, and whether the number is Armstrong or not.

Q5. Pattern Practice

Write programs for:
(a) Hollow square pattern (size N)
(b) Right triangle using numbers instead of stars

Q6. Loop Control Program

Print numbers from 1 to 40 with rules:
Skip multiples of 4.
Print 'Lucky' when number divisible by 7.
Stop when number reaches 29.

Section C

Q7. List Analysis

Take 10 numbers into a list and display:

all numbers, count of numbers greater than 50, average of list, and largest number.

Q8. String Analyzer

Take a sentence and display:

total characters (excluding spaces), total vowels, and total words.

Q9. Safe Division Program

Take two numbers and perform division.

Handle invalid input and division by zero.

Always print 'Program Ended' using finally.

Section D — Mini Project

Q10. Employee Management System

Create a menu-driven program using dictionary with initial data:

```
employees = { 'Amit': {'salary': 30000, 'dept': 'IT'}, 'Neha': {'salary': 28000, 'dept': 'HR'} }
```

Menu:

1 → Add employee, 2 → Update salary, 3 → Remove employee,

4 → Show all employees, 5 → Show highest salary employee, 6 → Exit.

Handle employee not found.